

4 (a) Outline the theory of evolution.

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(b) Approximately 95% of bird species living today belong to an evolutionary group known as Neoaves.

Bird species in the Neoaves group are believed to have evolved from a common ancestor that existed around 67 million years ago. The Neoaves that exist today can be divided into four groups:

- Mirandornithes, referred to as group 1
- Columbimorphae, referred to as group 2
- Otidimorphae, referred to as group 3
- all other Neoaves, referred to as group 4.

In 2014, scientists used DNA sequence data from 48 bird species to produce an evolutionary (phylogenetic) tree for the Neoaves. This is shown in Figure 4.1. The evolutionary tree shows possible evolutionary relationships between the four groups of species within the Neoaves.

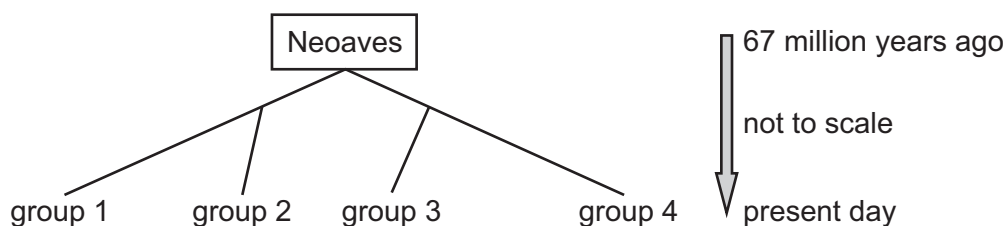


Figure 4.1

In 2024, scientists used DNA sequence data from 363 bird species to produce a new evolutionary tree for the four groups of species within the Neoaves. This is shown in Figure 4.2.

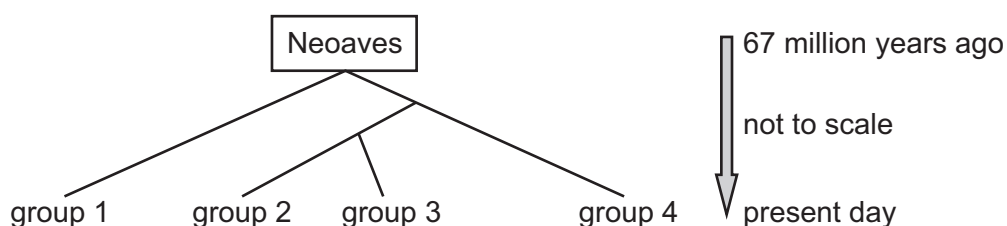


Figure 4.2



- (i) Describe the similarities and differences between the evolutionary relationships shown in Figure 4.1 and Figure 4.2.

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- (ii) Explain how DNA sequence data can be used to produce evolutionary trees such as those shown in Figure 4.1 and Figure 4.2.

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- (iii) State why Figure 4.2 is expected to be a more accurate representation of the evolutionary relationships of birds than Figure 4.1.

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- (c) During the evolution of some Neoaves species, crossing over during meiosis did **not** occur in a large section of one chromosome for several million years.

State **and** explain the effect on genetic diversity of an absence of crossing over.

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[Total: 12]